

Progress Report 1: Food Macro Estimation from Cooking Videos

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1 What Has Been Achieved

The goal of this project is to build an unsupervised computer vision pipeline that identifies food ingredients from cooking video frames and estimates their nutritional macronutrients using clustering-based classification and USDA FoodData Central mappings.

I. Dataset curation and organization. Six public image datasets were sourced, downloaded, and consolidated into a unified directory structure spanning **6 food classes** and **20 subclasses** (Table 1). Automated Python scripts handle the full pipeline: extraction of relevant subclasses from raw downloads, randomized 70/15/15 train/val/test splitting, JSON config generation, and visual summaries.

Class	Subclasses	Train	Source
Fruits	apple, banana, lemon, strawberry, orange	5,994	Fruits-360
Vegetables	onion, tomato, potato, carrot, broccoli	2,451	VegFru
Rice	arborio, basmati, ipsala, jasmine, karacadag	1,750	Rice Image Dataset
Seafood	shrimp, salmon, crab	892	Large Scale Fish
Oil	oil	685	Oil Detection
Egg	egg	113	Eggs Classification
Total		11,885	

Table 1: Dataset summary (train split). Rice capped at 500 images per variety to control class imbalance.

II. CLIP embedding extraction and unsupervised clustering. All 11,885 training images were encoded into 512-dimensional embeddings using OpenAI’s pretrained CLIP ViT-B/32—a Vision Transformer trained on 400M image-text pairs. No fine-tuning was performed. K-Means clustering ($k=6$ for classes, $k=20$ for subclasses) was applied to the embedding space (Table 2).

Level	ARI	NMI	Accuracy
Class ($k=6$)	0.8148	0.8827	98.35%
Subclass ($k=20$)	0.6614	0.8628	85.06%

Table 2: Clustering evaluation. ARI = Adjusted Rand Index; NMI = Normalized Mutual Information.

At the class level, fruits, rice, vegetables, and seafood achieved near-perfect separation ($F1 \geq 0.98$). Oil clustered well ($F1 = 0.91$) but absorbed all 113 egg images, resulting in 0.00 F1 for egg—attributable to small dataset size and visual similarity. At the subclass level, visually distinctive items (apple, strawberry, carrot, onion) achieved $F1 \geq 0.98$, while rice varieties could not be separated due to near-identical grain appearances. PCA and t-SNE visualizations confirm well-separated class clusters in the CLIP embedding space.

[GitHub Repository](#)

2 Immediate Next Steps

- Video frame pipeline.** OpenCV-based frame extractor that samples cooking video frames, embeds each via CLIP, and assigns to the nearest cluster centroid.
- USDA macro lookup.** Map detected classes to nutritional values (calories, protein, carbs, fat) and produce per-recipe macro estimates.

3. **LLM integration.** Generate natural-language nutritional summaries from detection output.

3 Challenges and Adjustments

1. **Unsupervised over supervised.** Clustering (K-Means on CLIP embeddings) was chosen over supervised classification (e.g., fine-tuning a CNN) because several sourced datasets were unlabeled, and manually annotating thousands of images was infeasible. The unsupervised approach also generalizes better—new food classes can be added by including reference images and re-running K-Means without retraining. This aligns with course themes: nearest-centroid assignment is search in embedding space, and cluster membership forms a probabilistic decision boundary.
2. **Dropped classes.** The original proposal included 9 classes (adding bread, butter, meat). These were removed because no large-scale single-item datasets exist for bread loaves/slices or butter sticks, and meat-type datasets are fragmented. Maintaining 6 well-supported classes was preferred over 9 poorly-supported ones.
3. **Egg absorption.** With only 113 images, egg was absorbed into the oil cluster. Mitigations include collecting more egg images (500+), using DBSCAN, or increasing k .
4. **Rice indistinguishability.** Rice subclasses (arborio, jasmine, karacadag) are visually identical. This is an inherent limitation of image-only grain classification, but acceptable since all rice varieties share nearly identical nutritional profiles.

4 Completion Plan

Mar 7–13 (<i>done</i>)	Dataset curation, train/val/test split, CLIP embedding extraction, K-Means clustering, evaluation (98.35% class accuracy)
Mar 14–20	Video frame extraction pipeline, nearest-centroid classification of frames, USDA macro lookup table
Mar 21–27	End-to-end integration, LLM nutritional summary, demo on real cooking video, test set evaluation
Mar 28–30	Final report writing, code cleanup, ablation studies (k sensitivity)